

Hepatitis viruses

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Lecture outline

- Objectives
- Hepatitis viruses :
 - Overview
 - Pathogenesis ,
 - Clinical features
 - Laboratory diagnosis
 - Management
- Summary
- Question

Objectives:

- Define hepatitis and viral hepatitis
- Different types of viral hepatitis
- Hepatitis viruses
 - The scope of the disease
 - Signs/symptoms
 - Diagnoses
 - Management

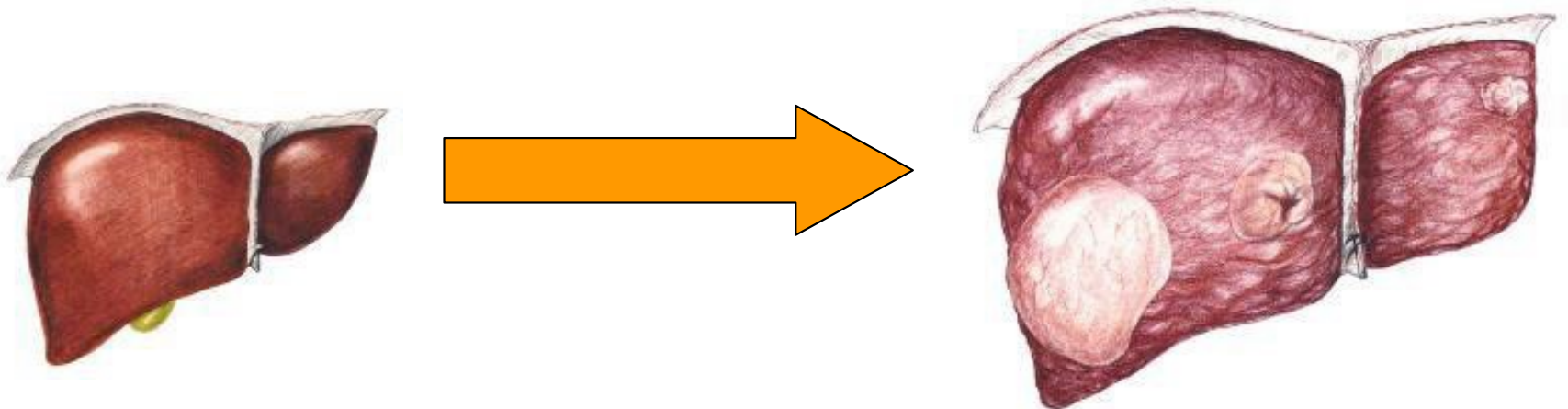
A middle-aged man complained of acute onset of fever, nausea, and pain in the right upper abdominal quadrant. There was jaundice, and dark urine had been observed several days earlier.

A laboratory test was positive for HAV IgM antibody.

- What will be the diagnosis?
 - How you advice the patient not to infect others ?
 - What's the main stay of management?
 - How you prevent the infection to high risk group?
-
- He may transmit the infection to family members by person-to-person spread for up to 2 weeks.. True or false
-
- The causative agent contains a single-stranded RNA genome. True or false

What Is Hepatitis?

- Hepatitis means inflammation of the liver
 - Hepat (liver) + itis (inflammation)= Hepatitis
- Viral hepatitis means there is a specific virus that is causing liver to inflame (swell or become larger than normal)



Causes of viral hepatitis

Common

Hepatitis A

Hepatitis B +/- D

Hepatitis C

Hepatitis E

Less common

Cytomegalovirus

Epstein-Barr virus

Rare

Herpes simplex

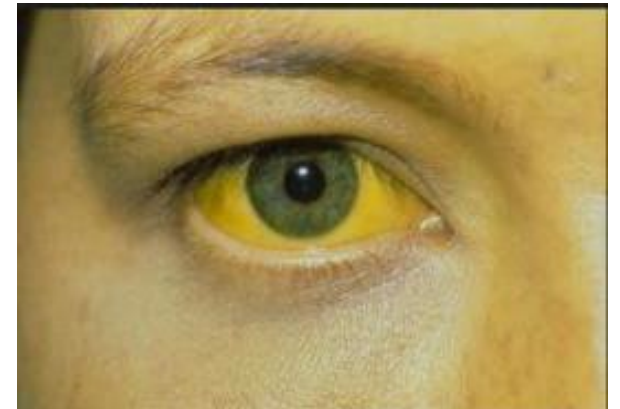
Yellow fever

Signs and Symptoms

- All these viruses cause similar clinical and pathological features, which are frequently anicteric or even asymptomatic.
- They may have one or more of the following symptoms:
 - Fever
 - Head ache
 - myalgia
 - arthralgia
 - Loss of appetite

Signs and Symptoms

- A few may have specific liver related symptoms initially:
 - Pale color stool
 - Dark color urine
 - Jaundice (yellowing of the skin or eyes)
 - Abdominal pain
 - Mild hepatosplenomegaly
 - Weight loss



Mild jaundice + abnormal liver blood tests
with nonspecific symptoms

Types of Hepatitis viruses:

Family

Hepatitis A (HAV): Picornoviridae family

Hepatitis B (HBV): Hepadnaviridae family

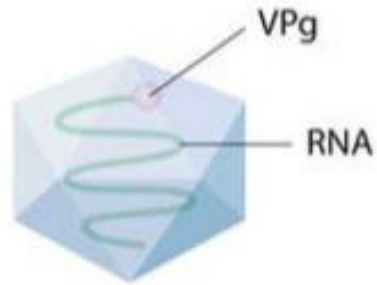
Hepatitis C (HCV): Flaviviridae

Hepatitis D (HDV): ? Delta virus

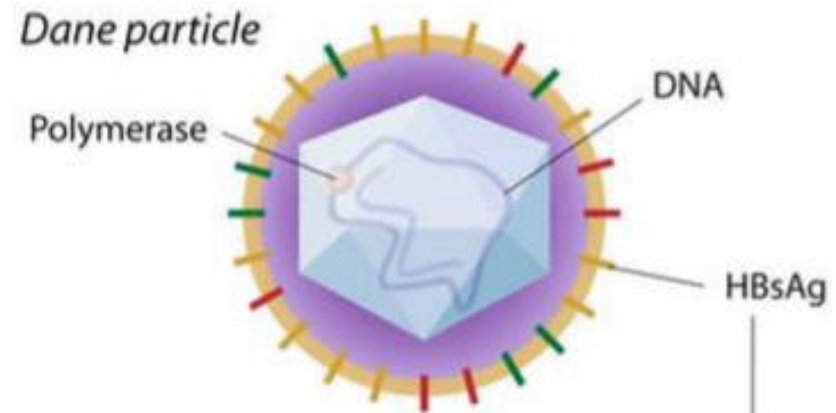
Hepatitis E (HEV): Hepeviridae

Hepatitis F (HFV): mutant of HBV

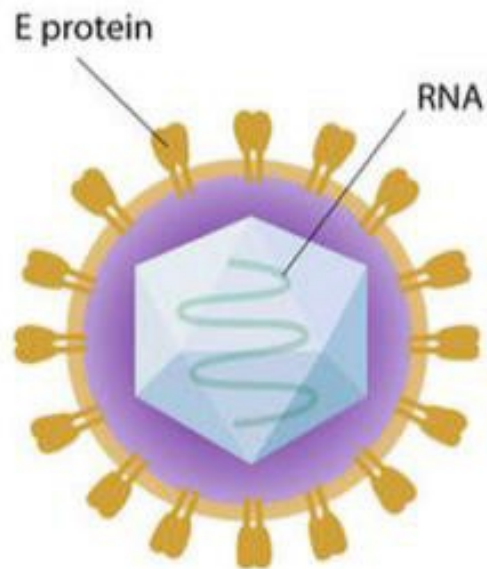
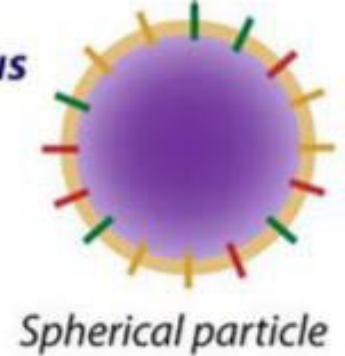
Hepatitis G (HGV): Flaviviridae



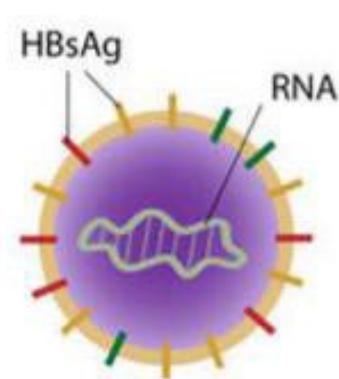
Hepatitis A vir



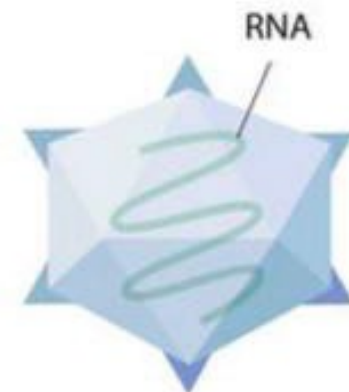
Hepatitis B virus



Hepatitis C virus



Hepatitis D virus



Hepatitis E virus

Types of Hepatitis viruses:

Genome

Hepatitis A (HAV): ssRNA

Hepatitis B (HBV): dsDNA

Hepatitis C (HCV): ssRNA

Hepatitis D (HDV): ssRNA

Hepatitis E (HEV): ssRNA

Types of Hepatitis viruses:

Structural composition

Hepatitis A (HAV): Naked

Hepatitis B (HBV): Enveloped

Hepatitis C (HCV): Enveloped

Hepatitis D (HDV): Enveloped

Hepatitis E (HEV): Naked

Types of Hepatitis viruses:

Ways of transmission

Hepatitis A (HAV): Enteric (Feco-Oral)

Hepatitis B (HBV): Parenteral

Hepatitis C (HCV): Parenteral

Hepatitis D (HDV): Parenteral

Hepatitis E (HEV): Enteric (Feco-Oral)

Types of Hepatitis viruses:

Stability

Hepatitis A (HAV): Heat & Acid Stability

Hepatitis B (HBV): Acid Sensitivity

Hepatitis C (HCV): Acid Sensitivity

Hepatitis D (HDV): Acid Sensitivity

Hepatitis E (HEV): Heat & Acid Stability

Types of Hepatitis viruses:

Incubation period/ weeks

Hepatitis A (HAV): 2-4

Hepatitis B (HBV): 4-20

Hepatitis C (HCV): 2-26

Hepatitis D (HDV): 6-9

Hepatitis E (HEV): 3-8

Types of Hepatitis viruses:

Prevalence

Hepatitis A (HAV): High

Hepatitis B (HBV): High

Hepatitis C (HCV): Moderate

Hepatitis D (HDV): Low regional

Hepatitis E (HEV): Regional

Types of Hepatitis viruses:

Chronicity

Hepatitis A (HAV): Never

Hepatitis B (HBV): Often

Hepatitis C (HCV): Often

Hepatitis D (HDV): Often

Hepatitis E (HEV): Never

Types of Hepatitis viruses:

Oncogenicity

Hepatitis A (HAV): Non-oncogenic

Hepatitis B (HBV): Oncogenic

Hepatitis C (HCV): Oncogenic

Hepatitis D (HDV): Unknown

Hepatitis E (HEV): Non-oncogenic

TABLE 41-2 Important Properties of Hepatitis Viruses

Virus	Genome	Replication Defective	DNA Polymerase in Virion	HBsAg in Envelope	Virus Family
HAV	ssRNA	No	No	No	Picornavirus
HBV	dsDNA ¹	No	Yes	Yes	Hepadnavirus
HCV	ssRNA	No	No	No	Flavivirus
HDV	ssRNA ²	Yes	No	Yes	Deltavirus
HEV	ssRNA	No	No	No	Calicivirus

Hepatitis A (Infectious hepatitis)

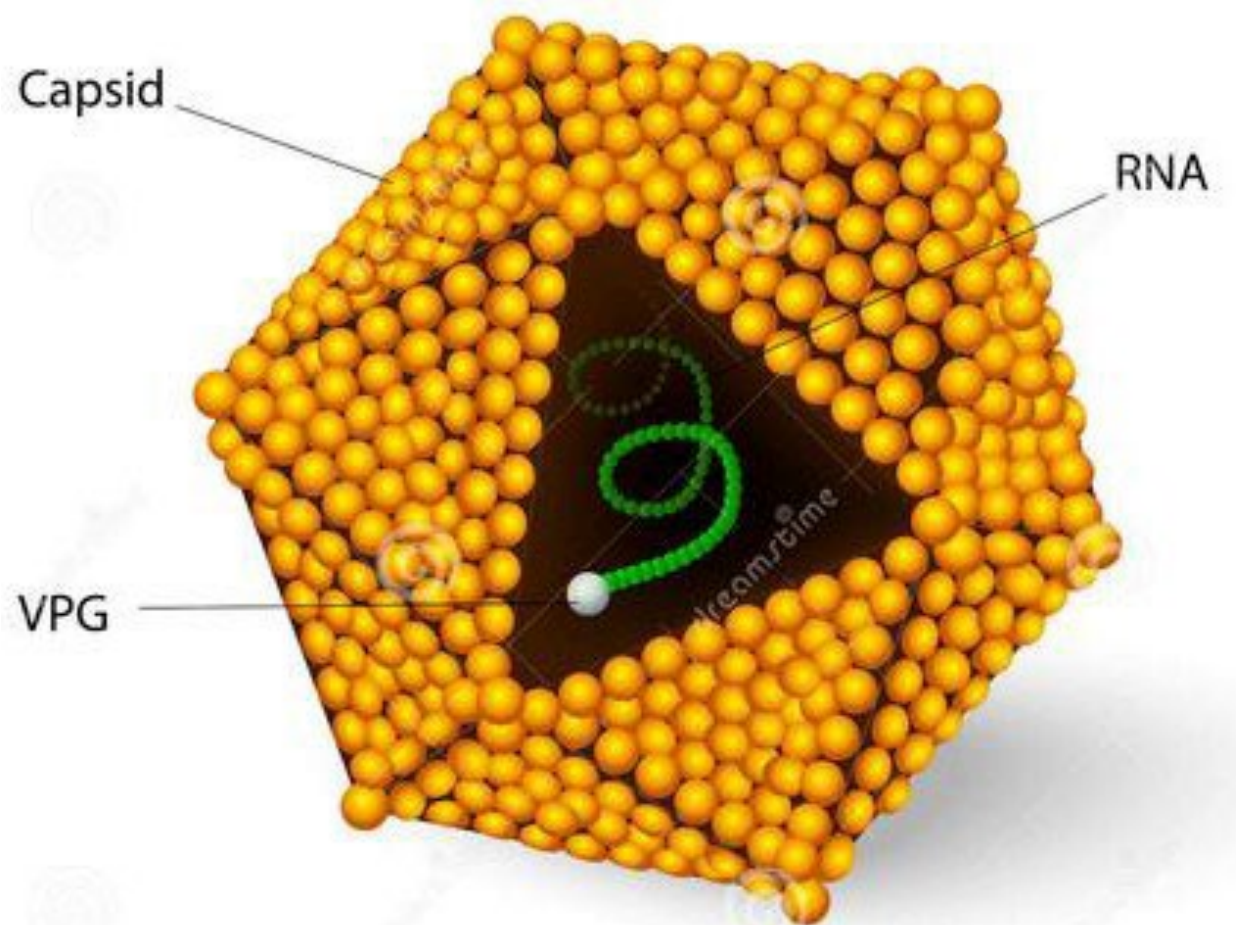


TABLE 36-1 Important Properties of Picornaviruses

Virion: Icosahedral, 28–30 nm in diameter, contains 60 subunits
Composition: RNA (30%), protein (70%)
Genome: Single-stranded RNA, linear, positive sense, 7.2–8.4 kb in size, molecular weight 2.5 million, infectious, contains genome-linked protein (VPg)
Proteins: Four major polypeptides cleaved from a large precursor polyprotein. Surface capsid proteins VP1 and VP3 are major antibody-binding sites. VP4 is an internal protein.
Envelope: None
Replication: Cytoplasm
Outstanding characteristic: Family is made up of many enterovirus and rhinovirus types that infect humans and lower animals, causing various illnesses ranging from poliomyelitis to aseptic meningitis to the common cold.

TABLE 36-2 Characteristics of Human Picornaviruses

Property	Human Enteroviruses A–D					Human Rhinoviruses A–C ^c	Human Parechoviruses ^d
	Polio	Coxsackie A ^a	Coxsackie B	Echo ^a	Entero ^b		
Serotypes	1–3	1–24	1–6	1–33	68–116	>150	1–14
Acid pH (pH 3.0)	Stable	Stable	Stable	Stable	Stable	Labile	Stable
Density (g/mL)	1.34	1.34	1.34	1.34	1.34	1.4	
Optimal temperature for growth (°C)	37	37	37	37	37	33	37
Common sites of isolation from humans							
Nose	0	0	0	0	0	+	0
Throat	+	+	+	+	+	+	
Lower intestine	+	+	+	+	+	0	+
Infect newborn mice ^e	0	+	+	0		0	

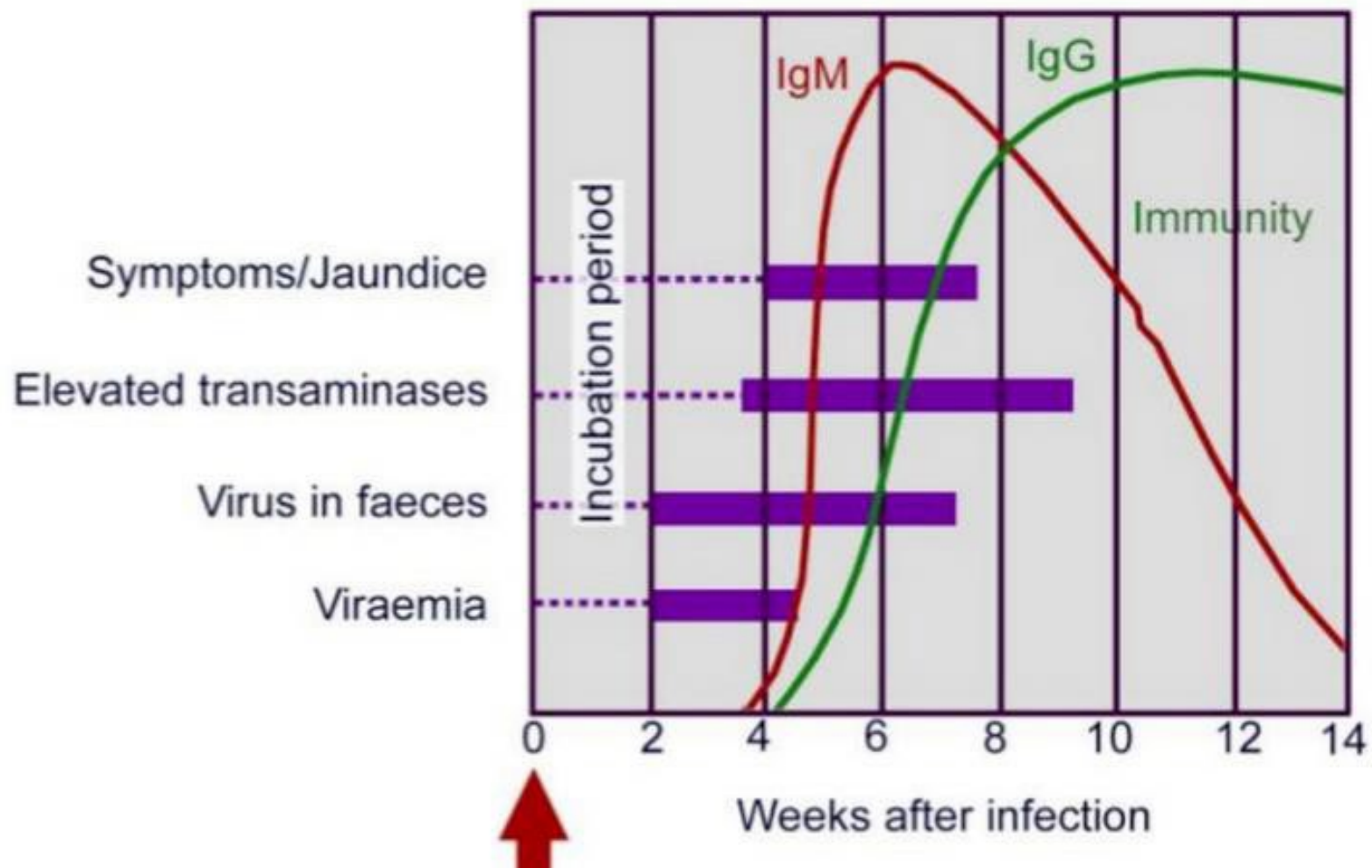
Hepatitis A

- Picornavirus group of enteroviruses
- Highly infectious
- Feco-oral
- Common in area of over crowding and poor sanitation
- The virus resistant to heat and chemicals.
- The disease is benign with complete recovery in several weeks.
- Immunity after HAV infection is usually ?

Hepatitis A /outcome

Outcome	Children	Adults
Inapparent (subclinical) infection (%)	80–95	10–25
Icteric disease (%)	5–20	75–90
Complete recovery (%)	>98	>98
Chronic disease (%)	None	None
Mortality rate (%)	0.1	0.3–2.1

Hepatitis A time course of infection



Hepatitis A

- **Diagnoses:**(serology),anti HAV antigen (IgM or IgG ?)
- **Prevention:** Better personal hygiene, passive immunization, active immunization
- **Treatment:** ?

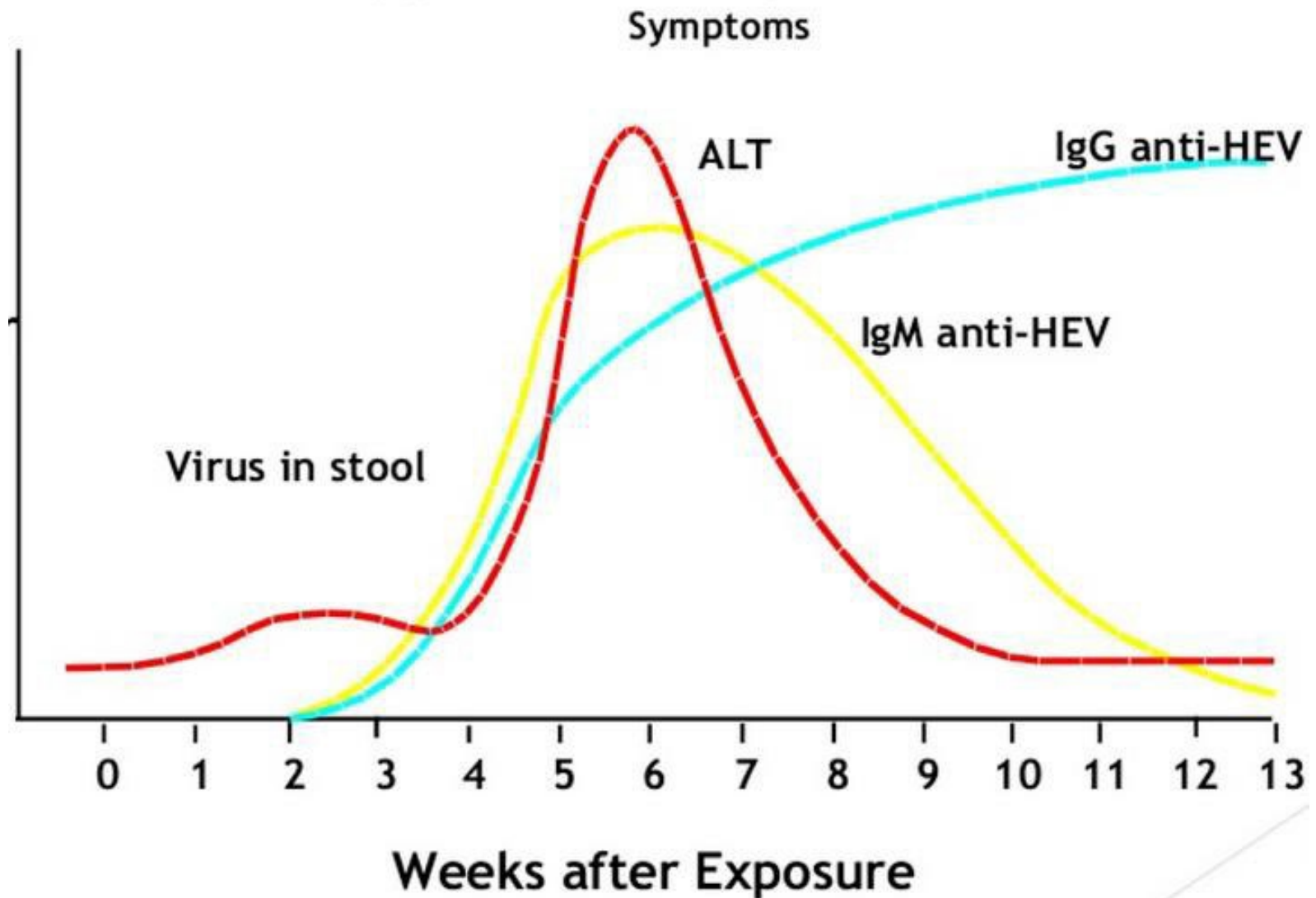
Hepatitis E (enteric NANB)

Hepatitis E

- Hepeviridae
- Feco-oral transmission .
- Endemic in India and Middle east
- Causes acute self-limiting infection with low mortality rate
- Higher mortality in ?
- Clinical presentation and management are similar to that of Hepatitis A

Hepatitis E

Typical Serologic Course



Hepatitis E

- Chronic carrier state does not occur
- Diagnoses:(serology) anti HEV antigen/IgM
- Prevention: Better personal hygiene, avoiding contaminated sea food.
- Treatment: supportive

Hepatitis E

Hepatitis E differs from hepatitis A in that infection during pregnancy is associated with development of acute liver failure which has a high mortality in acute infection

Questions ?